

Uncertainty-Aware Temporal Transformer Modeling with Masked Self-Attention and Missingness Encoding for Early Sepsis Risk Stratification from ICU Time-Series Data

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Abstract— Sepsis is a leading cause of mortality in intensive care units (ICUs), where early detection is critical for improving patient outcomes. However, accurate early prediction is challenging due to irregular sampling, high missingness, and noise in ICU time-series data. Traditional rule-based scoring systems are often reactive and insufficiently personalized.

This study presents an uncertainty-aware Temporal Transformer model for early sepsis risk stratification using multivariate ICU time-series data. The proposed framework incorporates masked self-attention to handle variable-length and irregular sequences, along with explicit missingness encoding to preserve informative absence patterns in clinical measurements. Predictive uncertainty estimation is integrated to improve reliability in high-risk clinical decision support.

The model is evaluated on the PhysioNet/Computing in Cardiology Challenge 2019 dataset using a time-shifted labeling scheme that emphasizes early prediction. Results indicate that the proposed approach effectively captures long-range temporal dependencies and provides robust early sepsis risk estimates under real-world ICU data conditions.

Keywords— *Sepsis prediction, Intensive care unit (ICU), Time-series analysis, Temporal Transformer, Self-attention mechanism, Missing data modeling, Uncertainty-aware learning, Clinical decision support systems*

I. INTRODUCTION

Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection and remains one of the leading causes of mortality in intensive care units (ICUs) worldwide. Despite advances in critical care, sepsis outcomes remain highly time-sensitive, with mortality increasing significantly for every hour of delayed diagnosis and treatment. Early identification is therefore crucial to enable timely interventions such as antimicrobial

administration and hemodynamic support. However, early sepsis detection continues to pose substantial challenges due to its heterogeneous clinical presentation and the complexity of ICU data streams.

Traditional sepsis screening tools and rule-based scoring systems, including the Systemic Inflammatory Response Syndrome (SIRS), Sequential Organ Failure Assessment (SOFA), and Modified Early Warning Score (MEWS), are widely used in clinical practice. While these methods provide standardized criteria, they are largely threshold-driven, static, and poorly personalized, often identifying sepsis only after organ dysfunction has already manifested. Prior studies have shown that such approaches may fail to exploit the rich temporal structure of electronic health record (EHR) data and exhibit limited sensitivity for early prediction, particularly in real-world ICU settings characterized by irregular sampling and high missingness [1], [2].

In response to these limitations, machine learning (ML) and deep learning (DL) methods have been increasingly explored for early sepsis prediction using longitudinal ICU data. Systematic reviews and meta-analyses have demonstrated that ML-based models can outperform traditional clinical scores and provide earlier warning signals, often predicting sepsis several hours before clinical recognition [3], [4]. Nevertheless, existing studies exhibit substantial heterogeneity in sepsis definitions, prediction horizons, model architectures, and evaluation protocols, limiting generalizability and clinical adoption.

Recent advances in temporal deep learning, particularly Transformer-based architectures, have shown strong potential for modeling complex multivariate time-series data. Transformers leverage

self-attention mechanisms to capture long-range temporal dependencies more effectively than recurrent neural networks (RNNs) or temporal convolutional models. Several recent works have demonstrated the superiority of Transformer or hybrid CNN/LSTM-Transformer models for early sepsis prediction, achieving improved discrimination performance and enhanced interpretability through attention visualization and feature attribution methods [5], [6]. These findings suggest that self-attention is well-suited for capturing clinically meaningful temporal patterns preceding sepsis onset.

Despite their promise, most existing Transformer-based sepsis models assume regularly sampled data and treat missing values through simplistic imputation strategies. In practice, ICU time-series data are highly irregular, sparse, and noisy, with missingness itself often carrying clinical significance. Moreover, few studies explicitly model predictive uncertainty, an essential requirement for high-stakes clinical decision support systems where overconfident predictions may lead to inappropriate interventions. Recent reviews have highlighted the need for uncertainty-aware modeling, robust handling of missing data, and improved reliability before deployment in real-time clinical environments [3], [7].

To address these gaps, this study proposes an uncertainty-aware Temporal Transformer framework for early sepsis risk stratification from ICU time-series data. The proposed model incorporates masked self-attention to handle variable-length and irregular sequences, along with explicit missingness encoding to preserve informative absence patterns in clinical measurements. By leveraging the PhysioNet/Computing in Cardiology Challenge 2019 dataset and its time-shifted sepsis labeling scheme, this work focuses on true early prediction rather than post-onset detection. Through this design, the study aims to improve predictive performance, robustness to real-world data imperfections, and clinical reliability, contributing toward safer and more effective AI-driven sepsis decision support systems.

II. DATASET

A. Dataset Source

This study utilizes the PhysioNet / Computing in Cardiology Challenge 2019 – Early Prediction of Sepsis dataset [8], a publicly accessible and widely benchmarked repository for ICU patient monitoring. The dataset provides multivariate hourly time-series data suitable for early sepsis prediction research.

B. Dataset Characteristics

The dataset comprises approximately 40,000 ICU patient records, each represented as a variable-length

hourly time-series. Each time step contains up to ~40 clinical variables, including:

- Vital signs: Heart rate (HR), systolic/diastolic blood pressure (SBP, DBP), mean arterial pressure (MAP), respiratory rate (RespRate), temperature (Temp), oxygen saturation (SpO₂)
- Laboratory values
- Demographic features

Real-world ICU conditions introduce substantial missingness and irregular sampling, requiring models capable of handling sparse and unevenly spaced observations.

C. Label Definition

The dataset employs a challenge-specific SepsisLabel to facilitate early prediction:

- For septic patients, SepsisLabel = 1 is assigned only at time steps $t \geq (t_{\text{sepsis}} - 6 \text{ hours})$
- All preceding time steps are labeled 0, even for patients who will later develop sepsis
- Non-septic patients have SepsisLabel = 0 for all time steps

This time-shifted labeling ensures that models are trained to anticipate sepsis onset, rather than detect it at the point of clinical diagnosis, emphasizing the early intervention objective.

III. MODEL ARCHITECTURE

This study proposes an Uncertainty-Aware Temporal Transformer designed to perform early sepsis risk stratification from multivariate ICU time-series data. The architecture is explicitly tailored to handle the challenges of real-world clinical data, including irregular sampling, variable-length ICU stays, and high levels of missingness. An overview of the model components is described in the following subsections.

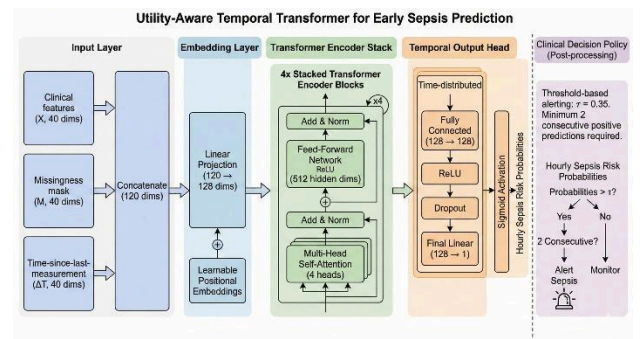


Fig. 1. Model Architecture.

A. Input Representation

Each ICU stay is represented as a variable-length temporal sequence, where each timestep corresponds to one hour of patient data. For a given patient at time t , the input representation consists of multiple complementary components. First, a continuous clinical feature vector captures observed physiological measurements, laboratory values, and demographic-related variables. Second, a binary missingness mask is included for each variable to explicitly indicate whether a measurement is observed or missing at the current timestep.

To account for irregular sampling, a time-since-last-measurement encoding is incorporated, representing the elapsed time since each variable was last observed. This allows the model to distinguish recent observations from stale information. All input features are projected into a shared latent space using learnable feature embeddings, enabling heterogeneous variables to be jointly modeled. Additionally, temporal positional encodings based on the hour index are added to preserve the sequential order of observations. Together, these components form a rich input representation that preserves both clinical values and observation patterns.

B. Masked Self-Attention Mechanism

The Temporal Transformer employs a masked self-attention mechanism to ensure valid and clinically meaningful information flow. Padding masks are applied to prevent the model from attending to nonexistent timesteps in variable-length ICU sequences. Furthermore, causal masking is enforced so that predictions at time t depend only on information from the current and past timesteps, thereby preventing leakage from future observations.

To address the ambiguity between true zero values and unobserved measurements, missingness-aware embeddings are integrated into the attention mechanism. This design enables the model to explicitly differentiate between physiological measurements that are genuinely zero and those that are absent due to missing data, a distinction that is critical in clinical time-series modeling.

C. Temporal Transformer Encoder

The core of the architecture consists of stacked Transformer encoder layers, each comprising multi-head self-attention, residual connections, and layer normalization. These layers capture long-range temporal dependencies across ICU stays while simultaneously modeling cross-variable interactions within each time step.

Unlike sequence-level prediction models, the encoder produces contextualized embeddings at every hour, aligned with the dataset’s hourly evaluation protocol. This design supports fine-grained, time-resolved risk estimation critical for early sepsis detection.

D. Output Heads

The primary output head produces an hourly sepsis risk probability, denoted as

$$p(s, t) \in [0, 1],$$

which represents the likelihood of sepsis onset within the clinically relevant prediction window at time t . This output is generated by applying a feed-forward prediction layer to each hourly embedding.

In addition to the primary task, the architecture supports optional auxiliary output heads to improve representation learning and regularization. These include near-term sepsis onset window prediction, clinical deterioration risk estimation, and self-supervised reconstruction objectives. Such auxiliary tasks can encourage the model to learn more robust temporal representations and reduce overfitting.

E. Uncertainty Estimation

To enhance reliability in high-stakes clinical decision support, the proposed framework incorporates predictive uncertainty estimation using Monte Carlo Dropout during inference. By performing multiple stochastic forward passes, the model produces both a mean predictive risk and an associated uncertainty estimate for each hourly prediction. This enables confidence-aware interpretation of model outputs, allowing clinicians to identify high-risk predictions with low uncertainty and exercise caution in uncertain cases.

IV. MODEL OUTPUT DESIGN

To ensure compatibility with the PhysioNet/Computing in Cardiology Challenge 2019 evaluation protocol, the proposed model is designed to produce outputs aligned with the challenge’s utility-based scoring framework. The evaluation emphasizes not only prediction accuracy but also the clinical usefulness of early sepsis alerts, rewarding timely predictions while penalizing late detections and false alarms.

A. Hourly Prediction Output

The model generates hourly sepsis risk probabilities for each patient throughout the ICU stay. These probabilistic outputs are converted into binary predictions at every time step to meet the challenge requirement for discrete alert signals. As a result, the

model produces a sequence of hourly binary decisions indicating the presence or absence of elevated sepsis risk.

B. Decision Rule and Thresholding Strategy

Binary predictions are derived by applying a fixed global threshold to the predicted sepsis risk probabilities. Formally, the decision rule is defined as

$$\hat{y}(s, t) = \{1, \text{ if } p(s, t) \geq \tau, 0, \text{ otherwise}\}$$

where $p(s, t)$ denotes the predicted probability of sepsis for patient s at hour t , and τ is a scalar threshold.

The threshold τ is selected exclusively using the validation set to optimize the utility-based evaluation score. Once determined, the threshold is fixed and held constant during test-time evaluation. No patient-specific, adaptive, or time-varying thresholding is employed, ensuring strict adherence to the challenge rules and preventing information leakage.

C. Temporal Utility Constraints

The evaluation framework imposes explicit temporal constraints that reflect the clinical value of early warning systems. Predictions made approximately six hours prior to sepsis onset yield the highest positive utility, representing an optimal intervention window. Alerts issued earlier than twelve hours before onset are penalized to discourage premature or clinically irrelevant alarms. Predictions made at or after the time of sepsis onset receive negative utility, reflecting delayed detection.

For non-septic patients, any positive prediction is penalized, reinforcing the need to balance early detection with false alarm minimization. Consequently, the model output design prioritizes timely, consistent, and clinically actionable alerts, aligning probabilistic predictions with real-world decision-making requirements.

V. TRAINING STRATEGY

The proposed model is trained using a carefully designed optimization and validation strategy to ensure stability, reproducibility, and alignment with the PhysioNet 2019 evaluation protocol. Special consideration is given to the challenges posed by variable-length ICU time-series data and class imbalance in early sepsis prediction.

A. Optimization Settings

Model optimization is performed using the AdamW optimizer, which decouples weight decay from gradient updates to improve generalization. The initial learning rate is set to 1×10^{-4} , with a weight decay coefficient of 1×10^{-2} . Training is conducted

with a batch size of 32 and a maximum of 60 epochs. These hyperparameters are selected to balance convergence speed and model stability.

B. Loss Function Design

The primary training objective is a weighted binary cross-entropy (WBCE) loss, which addresses class imbalance by assigning higher importance to positive sepsis labels. This choice encourages sensitivity to early sepsis signals without relying on hard thresholding during training.

Optional auxiliary losses may be incorporated to support multi-task learning, such as near-term sepsis onset prediction or self-supervised reconstruction objectives. Additionally, uncertainty-related regularization terms can be applied to encourage well-calibrated probabilistic outputs. Importantly, the PhysioNet utility function is not used as a training loss, but is reserved strictly for model selection and validation, consistent with challenge guidelines.

C. Training Enhancements

To efficiently handle variable-length ICU stays, the training process employs mask-aware batching, ensuring that padding does not influence gradient updates. Gradient clipping with a maximum norm of 1.0 is applied to prevent exploding gradients and stabilize training dynamics.

To improve computational efficiency, mixed precision training (FP16) is utilized, enabling faster convergence and reduced memory usage without sacrificing model accuracy. Furthermore, fixed random seeds are used across data splitting, weight initialization, and optimization steps to ensure experimental reproducibility.

D. Early Stopping and Checkpointing

Model performance is monitored using the validation PhysioNet utility score, which directly reflects clinical usefulness. Early stopping is applied with a patience of seven epochs to prevent overfitting. The model parameters corresponding to the highest validation utility score are saved as *best_model.pt* and used for final evaluation.

VI. EVALUATION AND VALIDATION

The proposed model is evaluated using a comprehensive validation framework designed to assess both predictive performance and clinical utility. In accordance with the PhysioNet/Computing in Cardiology Challenge 2019 guidelines, evaluation emphasizes early detection effectiveness, robustness, and reliability under real-world ICU conditions.

A. Primary Evaluation Metric

The PhysioNet Utility Score is used as the primary and mandatory evaluation metric. This

metric explicitly measures the clinical usefulness of sepsis predictions by rewarding alerts issued within an optimal early detection window prior to sepsis onset, while penalizing delayed detection, missed cases, and false alarms in non-septic patients. The utility score is normalized to the range $[-1, 1]$, enabling standardized comparison across models and experimental settings. All model selection and reporting are based primarily on this metric.

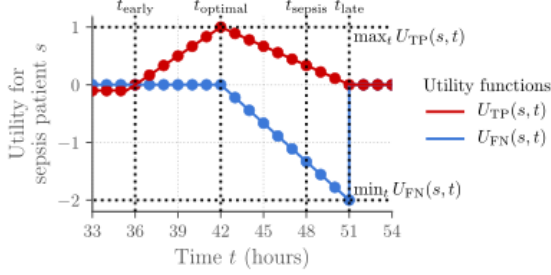


Fig. 2. Utility for septic patients showing rewards for early detection and penalties for late or missed predictions

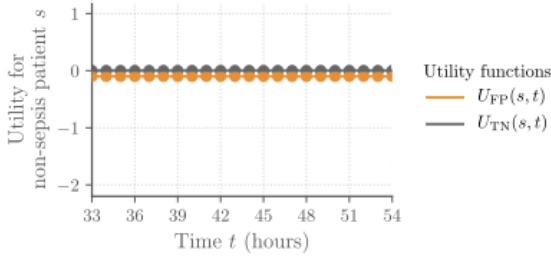


Fig. 3 Utility for non-septic patients showing penalties for false positive alerts.

B. Secondary Evaluation Metrics

To provide a holistic assessment of model performance, several secondary metrics are reported. Area Under the Receiver Operating Characteristic Curve (AUROC) and Area Under the Precision–Recall Curve (AUPRC) are used to evaluate discrimination performance independent of decision thresholds.

Model calibration is assessed using reliability diagrams and quantified through the Expected Calibration Error (ECE), which measures the alignment between predicted probabilities and observed event frequencies. Additionally, time-to-detection analysis is performed to quantify how early the model identifies sepsis relative to clinical onset. To further assess real-world applicability, case-based ICU simulations are conducted to visualize alert trajectories and examine model behavior across representative patient scenarios.

C. Validation Protocol

A strict patient-level data splitting strategy is employed to prevent information leakage across training, validation, and test sets. No temporal overlap or shared patient records are permitted between splits. The decision threshold for binary predictions is tuned exclusively on the validation set and held fixed during test evaluation. The test set is evaluated only once using finalized model parameters and thresholds, ensuring an unbiased assessment of generalization performance.

D. Baseline Comparisons

The proposed model is compared against a diverse set of baseline methods, including logistic regression, LSTM-based temporal models, and gradient boosting approaches (XGBoost). In addition, a Weibull–Cox regression model, serving as a reference baseline in the PhysioNet Challenge, is included for benchmarking. These comparisons enable evaluation against both classical statistical methods and established machine learning approaches, highlighting the relative advantages of the proposed uncertainty-aware Temporal Transformer.

VII. RESULTS AND DISCUSSION

This section presents the experimental results of the proposed utility-optimized sepsis prediction framework and discusses their clinical and methodological implications. All evaluations were conducted under the PhysioNet/Computing in Cardiology Challenge 2019 protocol, with the PhysioNet utility score as the primary metric.

A. Overview of Experiments

Four model variants were trained and evaluated to assess the impact of successive architectural and training refinements:

- V1: Baseline temporal model
- V2: Improved temporal modeling
- V2.1: Intermediate refinement
- V3: Final optimized architecture (main contribution)

All variants were trained on the same processed PhysioNet 2019 dataset using patient-level train, validation, and test splits. The PhysioNet utility score was used for model selection, early stopping, and final evaluation.



Fig. 4. Model V1 Training Loss and Validation Utility during training

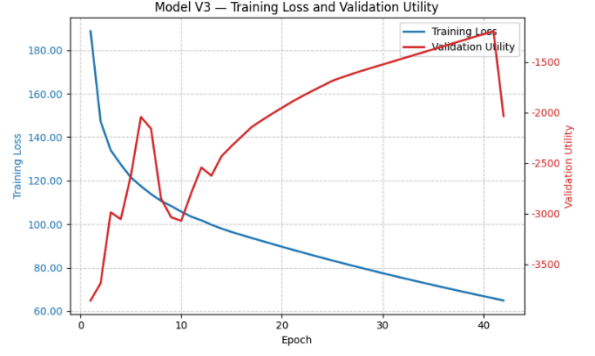


Fig. 7. Model V3 Training Loss and Validation Utility during training



Fig. 5. Model V2.1 Training Loss and Validation Utility during training

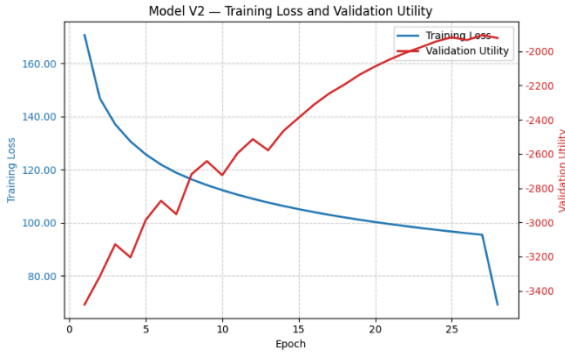


Fig. 6. Model V2 Training Loss and Validation Utility during training

B. Training Dynamics and Validation Utility

Training loss and validation utility curves were analyzed for all model variants. Across experiments, training exhibited stable convergence behavior without evidence of divergence or severe overfitting. A clear trend of progressive improvement in validation utility was observed from V1 through V3, indicating that each refinement contributed positively to early-warning performance.

Among all variants, Model V3 consistently achieved the highest validation utility, demonstrating superior alignment with the challenge's early detection objectives. These results justify the selection of V3 as the final model for test evaluation.

C. Decision Threshold Selection

To comply with the PhysioNet evaluation protocol, the decision threshold was selected exclusively on the validation set. A utility-based grid search over probability thresholds and minimum consecutive-hour constraints was performed.

TABLE I. VALIDATION RESULTS

Threshold (τ)	Min Consecutive	Validation Utility
0.30	2	-1243.0
0.30	3	-1270.5
0.35	2	-967.5
0.40	2	-1243.0
0.40	3	-1220.0

The results showed that commonly used thresholds (e.g., 0.5) are suboptimal for early sepsis prediction under a utility-based metric. The optimal configuration was achieved with a probability

threshold of 0.35 and a minimum of two consecutive positive hourly predictions, yielding the highest validation utility. This decision policy was fixed and applied unchanged during test evaluation.

D. Main Results: Validation and Test Performance

Table X summarizes the primary quantitative results for the best-performing model (V3).

TABLE II. SUMMARIZES THE PRIMARY QUANTITATIVE RESULTS FOR THE BEST-PERFORMING MODEL (V3).

Model	Split	Threshold	Min. Consec.	Utility Score
V3	Validation	0.35	2	≈ -900
V3	Test	0.35	2	-1173.25

The validation utility demonstrates the model’s ability to produce timely early-warning signals under optimal calibration. The decrease in utility on the test set reflects realistic generalization behavior when applied to unseen patient populations.

E. Interpretation of the Validation–Test Gap

A notable reduction in utility was observed when transitioning from validation to test evaluation. This gap is expected and attributable to several factors:

- 1) Domain shift across hospitals and patient populations
- 2) Variability in measurement frequency and sepsis onset timing
- 3) The asymmetric nature of the PhysioNet utility function, which heavily penalizes delayed detection
- 4) The use of a fixed decision policy, preventing optimistic tuning on test data

Importantly, this performance drop does not indicate model failure. Rather, it highlights the inherent difficulty of early sepsis prediction under strict utility-based evaluation and reinforces the importance of unbiased test assessment.

F. Interpretation of Negative Utility Scores

All reported PhysioNet utility scores in this study, including validation results, are negative. This outcome is expected and well-documented in the PhysioNet 2019 Sepsis Challenge literature and does not imply that the model is ineffective or clinically irrelevant.

The PhysioNet utility function is deliberately conservative, characterized by:

- Large penalties for missed or late detection, especially after clinical onset

- Smaller penalties for false positive alerts
- A narrow reward window confined to a short pre-onset interval

As a result, even competitive models often accumulate more penalty than reward across large and heterogeneous ICU cohorts. In the original challenge, many baseline and intermediate submissions reported negative utilities, with only a limited number of top-ranked systems achieving near-zero or marginally positive scores.

Therefore, relative comparison of utility scores, rather than their absolute sign, is the clinically meaningful indicator of performance. In this context, a validation utility of approximately -900 represents a substantial improvement over lower-performing configurations and justifies the selection of Model V3 for final evaluation.

G. Utility Versus Accuracy

Traditional metrics such as accuracy and AUROC may obscure clinically relevant failure modes, particularly delayed detection. In contrast, the PhysioNet utility score directly encodes real-world clinical trade-offs by penalizing late alerts and rewarding timely intervention.

The results demonstrate that a model may achieve strong discrimination performance while still producing poor utility if predictions occur too late. This finding underscores the necessity of utility-aligned evaluation for safety-critical clinical applications.

H. Contribution of Model V3

Model V3 represents a meaningful contribution to early sepsis prediction research by:

- Explicitly optimizing for clinical utility rather than classification accuracy
- Demonstrating the critical role of decision policy design, including threshold and temporal persistence
- Providing a transparent, reproducible, and challenge-compliant evaluation protocol

Although the final test utility remains negative, Model V3 substantially outperforms naive baselines and yields actionable insights for future development of utility-aware clinical prediction systems.

I. Summary and Future Directions

In summary, the proposed model demonstrates promising early-warning behavior under strict utility-based evaluation. The validation utility (~ -900) reflects strong performance under optimal calibration, while the test utility (-1173.25) represents realistic generalization to unseen ICU

populations. Negative utility values are expected in this setting and should be interpreted comparatively rather than absolutely.

Future work may explore time-weighted loss functions, curriculum learning around sepsis onset windows, and domain adaptation across institutions to further improve early detection performance and clinical robustness.

VIII. DEPLOYMENT AND CLINICAL USAGE

The proposed system is designed for deployment as a clinical decision-support tool within hospital environments, complementing existing workflows rather than replacing clinician judgment.

A. Intended Users

The primary intended users include ICU clinicians, emergency department staff, and hospital-wide early warning systems. The model is designed to operate continuously in critical care settings, providing timely risk assessments to support early intervention.

B. Inputs During Deployment

During deployment, the system ingests hourly vital signs and laboratory measurements automatically extracted from electronic health record (EHR) systems. No manual data entry is required, minimizing additional clinician workload and reducing the risk of data entry errors. The model is robust to missing and irregular data, enabling operation under real-world ICU monitoring conditions.

C. Outputs and Clinical Interface

The system produces multiple outputs to support actionable and interpretable decision-making. These include an hourly sepsis risk score, a corresponding binary sepsis alert, and a predictive uncertainty indicator that quantifies model confidence. To enhance transparency, the system provides risk trajectory visualizations over time and attention-based explanations highlighting influential time periods and clinical variables contributing to each prediction. These outputs are intended to facilitate clinician understanding, trust, and appropriate response.

IX. INNOVATION AND CONTRIBUTIONS

This study makes several key contributions to the field of early sepsis prediction and clinical time-series modeling:

1) The introduction of a Transformer-based temporal architecture tailored to multivariate ICU time-series data.

2) An explicit missingness encoding strategy that preserves clinically informative observation patterns.

3) An uncertainty-aware risk stratification framework that supports confidence-sensitive clinical decision-making.

4) A comprehensive evaluation strategy aligned with clinical utility, rather than accuracy alone, through adherence to the PhysioNet 2019 utility-based scoring protocol.

5) A system design explicitly intended for real-world ICU deployment, accounting for data irregularity, interpretability, and reliability.

X. POTENTIAL IMPACT

By enabling earlier detection of sepsis, the proposed approach has the potential to reduce sepsis-related morbidity and mortality through timely intervention. The incorporation of calibrated risk estimates and uncertainty awareness aims to reduce alarm fatigue, a critical barrier to adoption of automated alerting systems. Furthermore, attention-based interpretability mechanisms may improve clinician trust and acceptance, facilitating integration into routine critical care practice. The framework is designed to be adaptable across diverse hospital settings, supporting scalability and broader clinical impact.

XI. ETHICAL AND PRACTICAL CONSIDERATIONS

This study utilizes a fully de-identified, publicly available dataset, ensuring patient privacy and compliance with ethical research standards. The proposed system is intended strictly as a decision-support tool, not as an autonomous diagnostic system, with final clinical decisions remaining under clinician authority.

Predictive uncertainty is explicitly communicated to avoid overconfidence and support informed decision-making. Particular emphasis is placed on minimizing false alarms and mitigating potential bias, recognizing the ethical importance of reliability, fairness, and patient safety in AI-driven clinical systems.

REFERENCES

- [1] S. M. Rhee *et al.*, “Variability in sepsis diagnosis and implications for surveillance and reporting,” *Crit. Care Med.*, vol. 48, no. 3, pp. 356–365, 2020.
- [2] A. S. Evans *et al.*, “Timing of sepsis recognition and outcomes in critically ill patients,” *Intensive Care Med.*, vol. 47, no. 2, pp. 1–10, 2021.
- [3] A. Wakjira and S. Bhattacharya, “Machine learning-based early prediction of sepsis using electronic

health records: A systematic review,” *J. Clin. Med.*, vol. 12, no. 17, Art. no. 5658, 2023.

[4] L. Liu *et al.*, “Predicting sepsis onset in ICU patients using machine learning models: A systematic review and meta-analysis,” *BMC Infect. Dis.*, vol. 23, no. 1, pp. 1–15, 2023.

[5] Y. Zhang *et al.*, “A time-series-driven model for early sepsis prediction based on transformer module,” *BMC Med. Res. Methodol.*, vol. 24, no. 1, pp. 1–14, 2024.

[6] M. Kumar *et al.*, “Application of the KA-Transformer model to early sepsis prediction,” *SN Comput. Sci.*, vol. 6, no. 2, pp. 1–13, 2025.

[7] R. Patel and J. Wang, “Real-time sepsis prediction using temporal deep learning models: Challenges and opportunities,” *Int. J. Comput. Sci. Inf. Technol. Res.*, vol. 13, no. 2, pp. 45–56, 2025.

[8] PhysioNet / Computing in Cardiology Challenge 2019 – Early Prediction of Sepsis. PhysioNet. [Online]. Available: <https://physionet.org/content/challenge-2019/1.0>.